



Indian Institute of Technology
Kharagpur

S&P Global

CRISIL

TREMOR

Text-driven Risk Engine for Market Observation & Response

Converting unstructured financial news into explainable risk signals, market scenarios, and credit risk insights for faster, data-driven decisions.



Real-time
Event Understanding



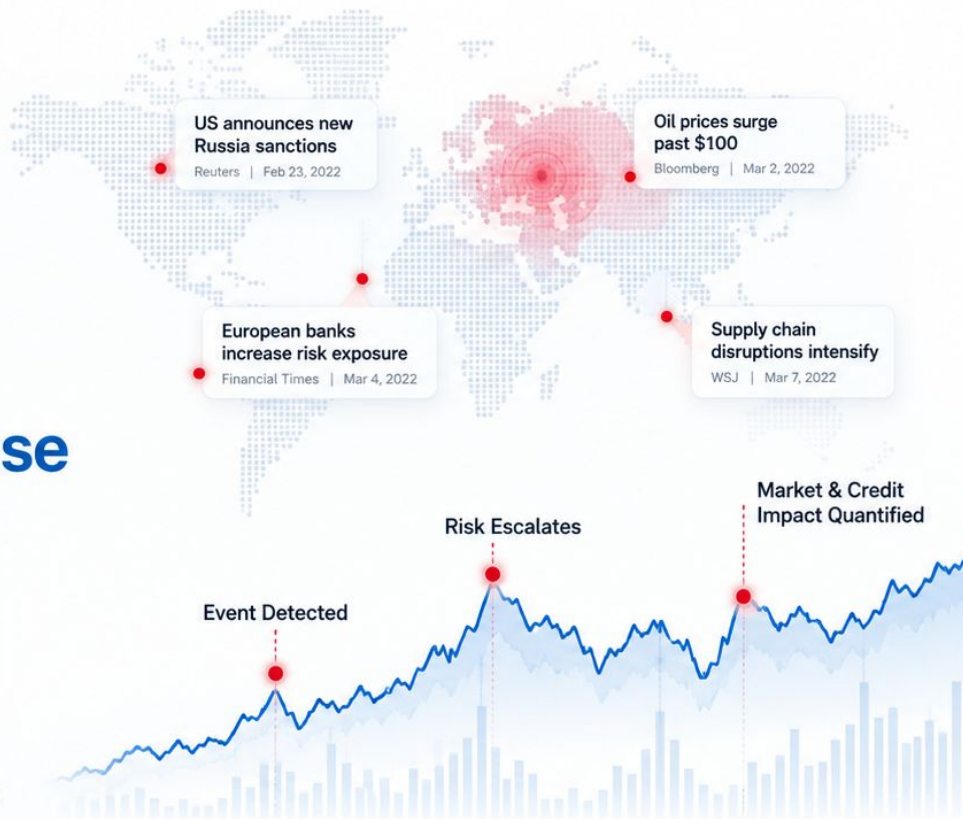
Actionable
Risk Signals



Market & Credit
Impact Analysis



Explainable
& Transparent





The Problem & Our Approach

Financial markets react to real-world events. The challenge is to convert the massive, unstructured news flow into timely, explainable and actionable risk insights.



The Problem

Global events, policy decisions and supply chain shocks move markets, but the information is unstructured, noisy and overwhelming.

A constant firehose of unstructured news

Reuters Feb 23, 2022
US announces new Russia sanctions
Washington unveils fresh measures targeting Russian banks and energy sector...

Financial Times Mar 4, 2022
European banks increase risk exposure
Lenders face higher counterparty and market risks amid escalating conflict...

WSJ Mar 7, 2022
Supply chain disruptions intensify
Global supply chains face renewed pressure as trade routes are disrupted...

⋮

Key challenges for risk teams



High volume & noise
Tens of thousands of news articles daily across sources, languages and regions.



Slow and manual analysis
Analysts spend significant time reading, tagging and interpreting events.



Hard to quantify impact
Linking events to market and credit impact requires domain expertise and structured analysis.



Lack of transparency
Black-box signals are hard to trust and act on in regulated environments.



Our Approach – TREMOR

Convert unstructured news into explainable risk signals, market scenarios, and credit risk insights.





TREMOR - System Architecture

From unstructured news and social media to structured risk signals, consumed via a tactical index, strategic stress test and a credit early-warning watchlist.

1

Data Sources

Unstructured real-time information



GDELT 2.0 (News)

global news, every 15 minutes
raw GKG files, no API key



RSS (News)

Yahoo Finance per ticker,
Google News topics, Federal Reserve



StockTwits + Bluesky (Social)

live cashtag streams, no API key;
replays; tweet archive, Hacker News



Replay Packs (History)

Russia-Ukraine, 21-24 Feb 2022;
SVB + bank contagion, 8-15 Mar 2023



Prices (Finance)

constituents, risk factors,
26 historical episodes

4

Offline: Training, Evaluation, Validation

public labelled data (PhraseBank, TFNs, FIQA, SntFIN, StockTwits self-labels, topic tweets) + weak labels (news searches, GDELT)
→ multi-task fine-tune (8 min, laptop GPU) with embedding distillation → ONNX int8 → held-out evaluation vs keyword baseline, FinBERT;
Module A backtest; point-in-time scenario backtest (21 crises); impact event study vs market moves; price-direction check

2

AI / NLP Risk Engine

Synchronous, deterministic, event-time clock – the same code runs live and on replays



1 Gate

drop text naming no tracked entity /
risk theme (~10× volume cut, zero model cost)



2 De-duplicate

fingerprint: syndicated copies,
corroboration, not new information



3 Entity Linking

88 entities: companies, banks, countries,
central banks, commodities; ambiguity +
context guards



4 One Encoder, Many Heads

fine-tuned bge-small, 34 MB, ~4 ms/headline:
embedding | sentiment | event type |
entity sentiment



5 Story & Event Clustering

single-pass, online: stories (same report),
events (related stories) → novelty



6 Impact Scorecard (1-10)

base severity + intensity + corroboration
+ velocity + breadth + market linkage



7 Entity Sentiment State

Bayesian filter with decay + uncertainty;
news vs social divergence



8 Signals

Doc / Event / Entity signals:
sentiment: -1..1, event type, impact 1-10,
price direction



Pub/Sub Bus → REST API (/docs) + Live Event Stream (SSE) + JSONL Signal File
in-process today; integration point where Kafka / Redis Streams would sit in production

3

Downstream Applications

Turn signals into actionable insights



Module A – Tactical Index Rebalancer (TREMOR-20)

20 S&P 100 stocks; weights rise with positive
sentiment and fall with negative.



Module B – Strategic Stress Test

event type + impact score + market correlations
to estimate portfolio losses (e.g., P&L, CET1 ratio).



Credit Early-Warning Watchlist

entity sentiment state points scorecard →
watch negative / monitor exposure; lead time
measured against public rating actions.

5

Dashboard (No Build Step)

radar · index · stress lab · credit watch · analyze ·
results; live via server-sent events; light / dark



Implementation Highlights – Core Components of TREMOR

A modular pipeline that converts unstructured news into risk signals, market scenarios and credit risk insights.

1 NLP Engine – From News to Structured Events

Detects and classifies financial events with context using a fine-tuned language model.



News Articles
(Reuters, FT, WSJ, ...)



Preprocessing
Cleaning, deduplication,
segmentation



Event Detection & Classification
(what, when, where, type)



Entity & Risk Factor Mapping
(sector, region, asset class)



Structured Events
(for downstream modules)

Event Type Classification (Macro F1)

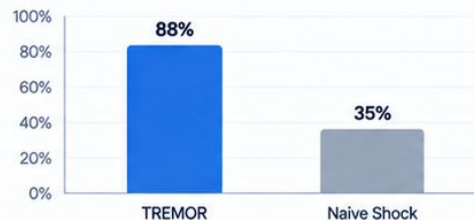


- 39,242 documents processed
- 307 docs/second
- 1.5 – 3 minutes total processing time
- ~4.2 ms per headline (CPU)
- 34 MB model size

2 TREMOR Index – Event-driven Risk Signals

Converts structured events into a time-series index capturing market risk factor movements.

TREMOR Index vs Naive Shock – Risk Factor Direction Accuracy



Captures key market risk factors

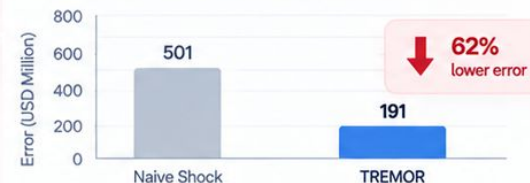
- Equity Markets
- Interest Rates
- Commodities
- Credit Spreads
- FX
- Volatility

3 Credit Stress Test – From Events to Portfolio Impact

Translates market scenarios into potential credit losses and capital impact for financial institutions.



Mean P&L Error vs Naive Baseline



Portfolio & Capital Impact (Ukraine Escalation Example)

Impact Score
7.7 → 9.9
(escalation)

CET1 Ratio
13.5% → 9.68%
(at peak)

Risk Factor Directions
8 / 9
correct



Technology Stack



Python
(Core pipeline)



Hugging Face
(NLP models)



PostgreSQL
(Data storage)



FastAPI
(API & services)



Streamlit
(Dashboard)



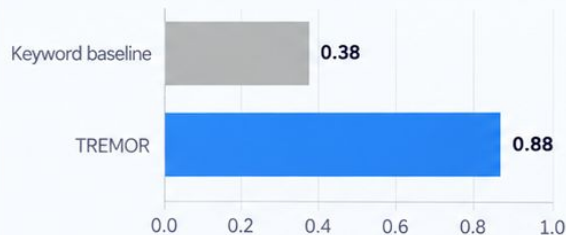
TREMOR against the Naive Approach

Held-out data and point-in-time history only. Every number comes from docs/results/*.json; stronger baselines and all metrics are in docs/RESULTS.md.

1

Event Type Classification

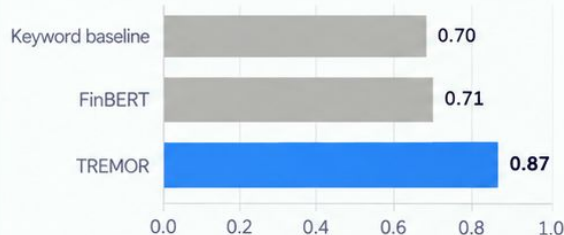
Macro-F1 on human-labelled test set



2

Tweet Sentiment

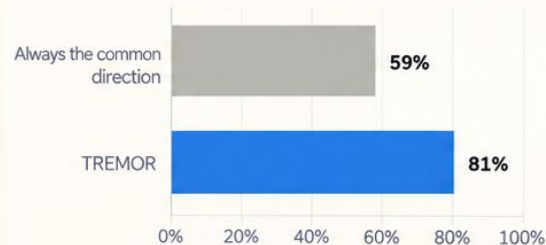
Accuracy on held-out financial tweets (TFNS)



3

Price Direction from Headlines

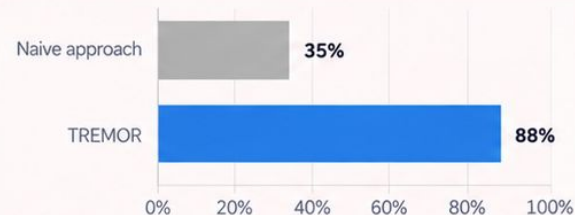
Agreement with the real move (175 headlines)



4

Stress Scenarios: Directions

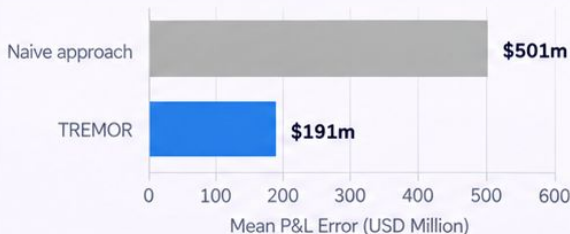
Risk-factor moves called right (21 past crises)



5

Stress Scenarios: P&L Error

Mean error in the bank's P&L (21 past crises) (lower is better)



6

Signal, Not Noise

The four-day Ukraine replay

39,242

documents ingested

14

stress tests out,
each tied to a real
development



Case Study: Russia-Ukraine Escalation (21-24 Feb 2022)

TREMOR captures real-world events in real time, with the risk index peaking on 24 Feb 2022 and correctly identifying key risk factors.

1

Event Timeline and TREMOR Index

Key events and TREMOR impact score (real time)



21 Feb 11:45
US moves to cut off
Russian banks

22 Feb 01:00
Recognition of
separatist regions

24 Feb 05:45
Putin announces military
assault against Ukraine
in surprise speech

2

Risk Factor Movements

(16 Feb – 8 Mar 2022)

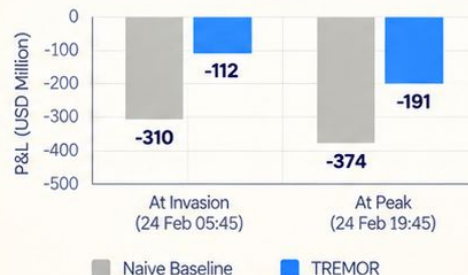
Direction predicted by TREMOR vs actual move

Risk Factor	TREMOR Predicted Move	Actual Move	Correct?
Oil Prices	↑	↑	✓
Natural Gas Prices	↑	↑	✓
Equity Markets	↓	↓	✓
Sovereign Spreads	↑	↑	✓
USD Index	↑	↑	✓
Interest Rates	↓	↓	✓
Credit Spreads	↑	↑	✓
Volatility (VIX)	↑	↑	✓
Gold Prices	↑	↑	✓
Euro (vs USD)	↑	↓	✗

3

Portfolio P&L at Key Moments

Cumulative losses for the representative portfolio



49% lower losses at peak
vs. naive baseline (-\$191M vs. -\$374M)

4

Credit Early-Warning Watchlist

Flagged names before rating actions (selected examples)

Company	Event Date	TREMOR Flag (Days Before)	Actual Downgrade
Sberbank	22 Feb 2022	7 days	24 Feb 2022
VTB	22 Feb 2022	6 days	24 Feb 2022
Gazprom	21 Feb 2022	5 days	24 Feb 2022
Sovcombank	22 Feb 2022	6 days	24 Feb 2022

5

Narrative → Signals

Example event to signal extraction

Headline

"Oil On The Boil: Brent
races to \$105 amid
Russia-Ukraine
hostilities"
24 Feb 2022, 09:20



Extracted Signals

Sentiment: -1.1
Event Type: Geopolitical
Impact Score: 9.4
Price Direction: ↑ (Oil)

6

Key Takeaway



- TREMOR detects and quantifies market-moving events in real time.
- Provides accurate risk signals and factor directions.
- Significantly lower portfolio losses vs. naive approach.
- Enabled early identification of credit risks.



Conclusion & Future Scope

TREMOR converts unstructured news and social media into actionable, quantifiable risk signals for financial markets.

1

Key Results

TREMOR outperforms naive baselines across tasks.



**Event Classification
(macro-F1)**
vs. keyword baseline (0.38)

0.88
(+132%)



**Tweet Sentiment
(accuracy)**
vs. FinBERT (0.71)

0.87
(+23%)



Price Direction (headlines)
vs. common direction (59%)

81%
(+22 pp)



Stress Test (21 past crises)
Directions predicted correctly

88%
vs. 35%



**Portfolio P&L Error
(mean over 21 crises)**
vs. naive baseline (\$501m)

\$191m
(↓ 62%)

2

Key Takeaways

What we learned from building and evaluating TREMOR.



Early and Accurate Detection

Captures market-relevant events in real time, providing early warning signals (e.g., 3 days before peak in Ukraine case).



Structured and Interpretable Signals

Converts noisy news and social media into interpretable risk signals with quantifiable impact scores.



Consistent Outperformance

Outperforms keyword and language-model baselines across event classification, sentiment analysis and market impact prediction.



Practical for Risk Management

Useful for portfolio risk monitoring, credit watchlists and scenario analysis during global stress events.

3

Future Scope

Directions to further improve and extend TREMOR.



Broader Coverage

Extend to more asset classes, geographies and languages for global applicability.



Richer Models

Incorporate graph-based event propagation and multi-modal signals (audio, video).



Live Deployment

Scale real-time streaming pipeline for production use with continuous monitoring.



Enhanced Risk Applications

Integrate with portfolio optimization, stress testing and automated trading workflows.

4

Resources

Code and documentation for reference.



GitHub Repository

<https://github.com/ShubAce/IITKharagpur-Shubham-Suman-hackathon>

5

Team

Team DASK – IIT Kharagpur (Team ID: 178814)



Shubham Suman
Team Leader
IIT Kharagpur



Anurag Ranjan
Team Member
IIT Kharagpur



Krupal Thakre
Team Member
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Durgesh Kumar
Team Member
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